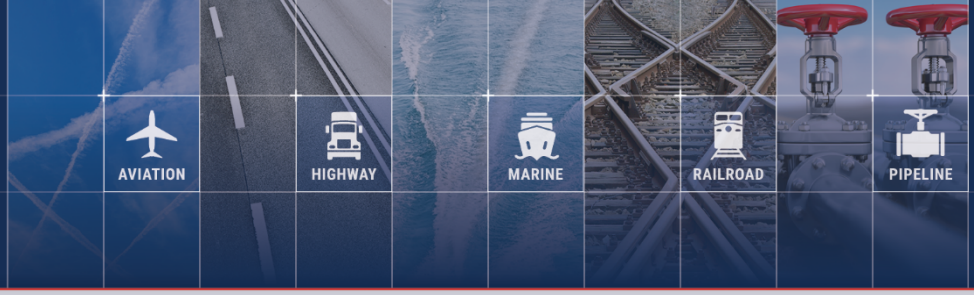

Aviation Investigation Final Report

Location:	New Braunfels, Texas	**Accident Number:**	CEN25LA065
Date & Time:	December 27, 2024, 22:45 Local	**Registration:**	N5263W
Aircraft:	Piper PA-28	**Aircraft Damage:**	Substantial
Defining Event:	Fuel starvation	**Injuries:**	2 Serious, 1 Minor
Flight Conducted Under:	Part 91: General aviation - Personal		

Analysis

Prior to the flight, the pilot checked the fuel level and noted the airplane contained about 38 gallons of fuel. During the cross-country flight, the pilot switched fuel tanks about every 15 minutes. While on approach about 3 miles from the destination airport with the fuel selector in the left tank position, the engine began to intermittently lose power. With the electric fuel pump on, the pilot pumped the throttle, and the engine temporarily regained full power. The pilot attempted to gain some altitude when the engine lost total power. Unable to make the airport, the pilot conducted an off-field landing, during which the left wing sustained substantial damage. Postaccident examination of the airplane revealed that the left fuel tank was empty, and the right fuel tank contained 8 gallons of fuel.

Probable Cause and Findings

The National Transportation Safety Board determines the probable cause(s) of this accident to be:

The pilot’s improper in-flight fuel planning which resulted in a total loss of engine power due to fuel starvation and a subsequent off-field landing on rough terrain.

Findings

Personnel issues	Fuel planning - Pilot
Aircraft	Fuel - Fluid level
Environmental issues	Rough terrain - Contributed to outcome

Factual Information

History of Flight

Approach-VFR pattern final	Fuel starvation (Defining event)
Landing-flare/touchdown	Off-field or emergency landing

Pilot Information

Certificate:	Private	**Age:**	28,Male
Airplane Rating(s):	Single-engine land	**Seat Occupied:**	Left
Other Aircraft Rating(s):	None	**Restraint Used:**	Lap only
Instrument Rating(s):	None	**Second Pilot Present:**	No
Instructor Rating(s):	None	**Toxicology Performed:**	
Medical Certification:	Class 1 Without waivers/limitations	**Last FAA Medical Exam:**	November 3, 2022
Occupational Pilot:	No	**Last Flight Review or Equivalent:**	August 20, 2024
Flight Time:	118.4 hours (Total, all aircraft), 3.5 hours (Total, this make and model), 50.7 hours (Pilot In Command, all aircraft), 16.4 hours (Last 90 days, all aircraft), 15.3 hours (Last 30 days, all aircraft)		

Aircraft and Owner/Operator Information

Aircraft Make:	Piper	**Registration:**	N5263W
Model/Series:	PA-28	**Aircraft Category:**	Airplane
Year of Manufacture:	1962	**Amateur Built:**	
Airworthiness Certificate:	Normal	**Serial Number:**	28-309
Landing Gear Type:	Tricycle	**Seats:**	2
Date/Type of Last Inspection:	Unknown	**Certified Max Gross Wt.:**	2400 lbs
Time Since Last Inspection:		**Engines:**	1 Reciprocating
Airframe Total Time:		**Engine Manufacturer:**	Lycoming
ELT:		**Engine Model/Series:**	O-360
Registered Owner:	On file	**Rated Power:**	180 Horsepower
Operator:	On file	**Operating Certificate(s) Held:**	None

Meteorological Information and Flight Plan

Conditions at Accident Site:	Visual (VMC)	**Condition of Light:**	Night
Observation Facility, Elevation:	KBAZ,648 ft msl	**Distance from Accident Site:**	0 Nautical Miles
Observation Time:	22:51 Local	**Direction from Accident Site:**	336°
Lowest Cloud Condition:	Clear	**Visibility**	10 miles
Lowest Ceiling:	None	**Visibility (RVR):**	
Wind Speed/Gusts:	/	**Turbulence Type Forecast/Actual:**	/
Wind Direction:		**Turbulence Severity Forecast/Actual:**	/
Altimeter Setting:	29.83 inches Hg	**Temperature/Dew Point:**	16°C / 11°C
Precipitation and Obscuration:	No Obscuration; No Precipitation		
Departure Point:	Midlothian, TX (JWY)	**Type of Flight Plan Filed:**	VFR
Destination:	New Braunfels, TX	**Type of Clearance:**	None
Departure Time:	20:44 Local	**Type of Airspace:**	Class D

Wreckage and Impact Information

Crew Injuries:	1 Minor	**Aircraft Damage:**	Substantial
Passenger Injuries:	2 Serious	**Aircraft Fire:**	None
Ground Injuries:		**Aircraft Explosion:**	None
Total Injuries:	2 Serious, 1 Minor	**Latitude, Longitude:**	29.702707,-98.042741(est)

Administrative Information

Investigator In Charge (IIC):	Sauer, Aaron
Additional Participating Persons:	Keenon Wood; FAA; San Antonio, TX
Original Publish Date:	January 30, 2025
Last Revision Date:	
Investigation Class:	Class 4
Note:	The NTSB did not travel to the scene of this accident.
Investigation Docket:	https://data.ntsb.gov/Docket?ProjectID=199488

The National Transportation Safety Board (NTSB) is an independent federal agency charged by Congress with investigating every civil aviation accident in the United States and significant events in other modes of transportation—railroad, transit, highway, marine, pipeline, and commercial space. We determine the probable causes of the accidents and events we investigate, and issue safety recommendations aimed at preventing future occurrences. In addition, we conduct transportation safety research studies and offer information and other assistance to family members and survivors for each accident or event we investigate. We also serve as the appellate authority for enforcement actions involving aviation and mariner certificates issued by the Federal Aviation Administration (FAA) and US Coast Guard, and we adjudicate appeals of civil penalty actions taken by the FAA.

The NTSB does not assign fault or blame for an accident or incident; rather, as specified by NTSB regulation, "accident/incident investigations are fact-finding proceedings with no formal issues and no adverse parties … and are not conducted for the purpose of determining the rights or liabilities of any person" (Title 49 *Code of Federal Regulations* section 831.4). Assignment of fault or legal liability is not relevant to the NTSB's statutory mission to improve transportation safety by investigating accidents and incidents and issuing safety recommendations. In addition, statutory language prohibits the admission into evidence or use of any part of an NTSB report related to an accident in a civil action for damages resulting from a matter mentioned in the report (Title 49 *United States Code* section 1154(b)). A factual report that may be admissible under 49 *United States Code* section 1154(b) is available here.